



Project Title -:Car Market Trends Analysis with Car Dekho Data

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PROBLEM STATEMENT

- Determining key market valuation drivers, depreciation patterns, and pricing disparities in the secondary automotive market to establish transparent, data-driven used car resale pricing.



Project Description :-

Used vehicle resale values fluctuate heavily based on age, original showroom price (Present_Price), cumulative distance (Kms_Driven), fuel technology (Fuel_Type), sales channel (Seller_Type), and gearbox transmission (Transmission). In the CarDekho dataset of 301 listed vehicles, average selling prices drop from original showroom levels of ₹7.63 Lakh down to ₹4.66 Lakh, exhibiting significant variance across individual sellers (avg ₹0.87 Lakh) and registered dealers (avg ₹5.71 Lakh). Without quantitative pricing models, sellers risk mispricing vehicles, buyers face overpaying, and dealerships encounter inventory holding losses due to inaccurate valuation metrics.

WHO ARE THE END USERS?

- **Automotive Dealerships & Fleet Managers:** Optimizes trade-in valuations, price positioning, and profit margins on pre-owned vehicle stock.
- **Individual Vehicle Buyers & Sellers:** Provides benchmarks to negotiate fair market values based on vehicle age, mileage, and transmission type.
- **Used Car E-Commerce Platforms (e.g., CarDekho, Cars24):** Enhances automated valuation algorithms and price prediction engines for user listings.
- **Auto Financing & Insurance Underwriters:** Assesses asset depreciation rates and loan-to-value ratios for pre-owned car financing.

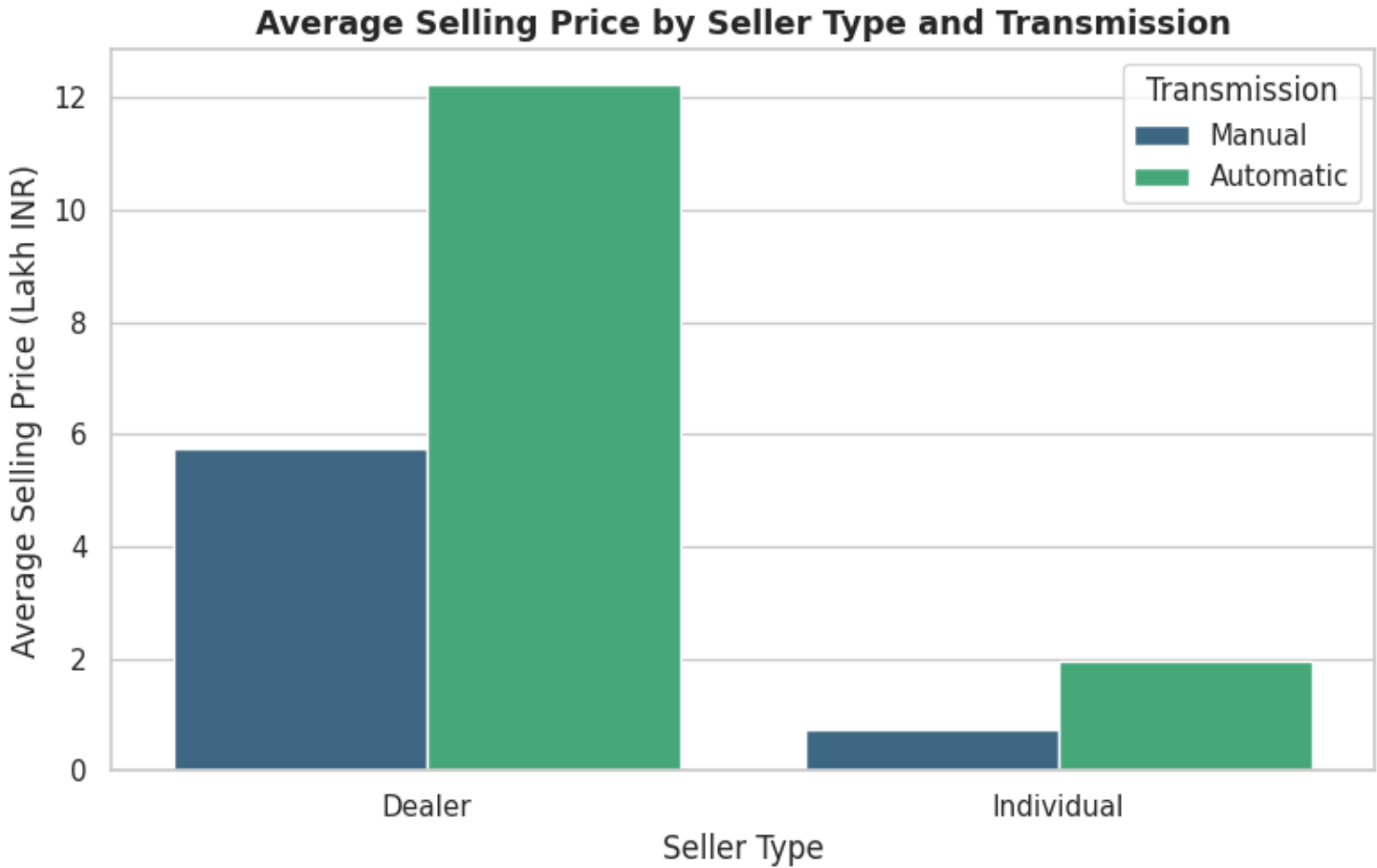
Technology Used

- Programming Language: Python (3.x)
- Data Manipulation & Processing: Pandas, NumPy
- Data Visualization: Seaborn, Matplotlib
- Execution Environment: Google Colab / Jupyter Notebook



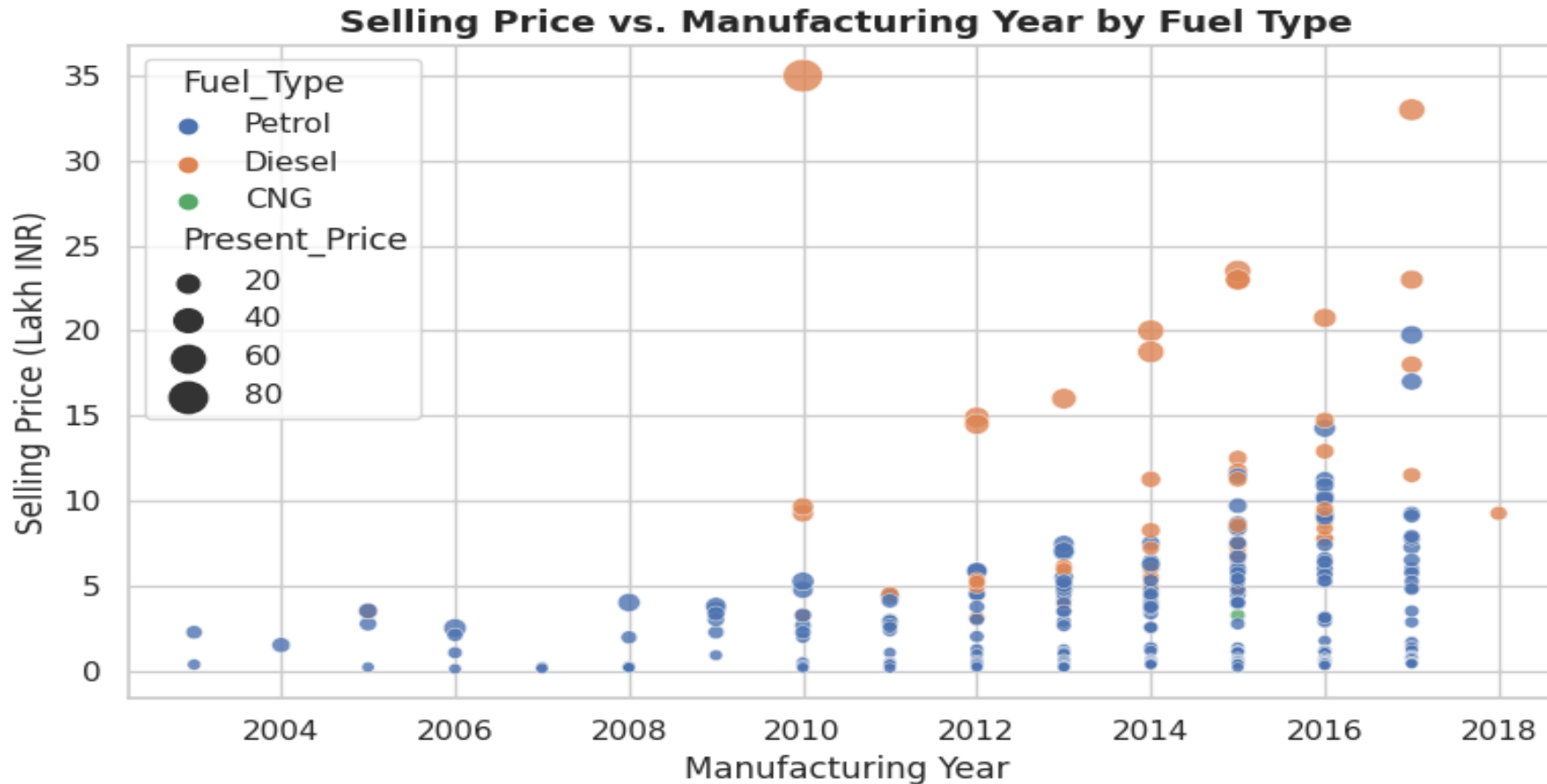
RESULTS

Average Selling Price Breakdown by Seller Channel and Transmission



RESULTS

Depreciation Trend & Selling Price by Manufacturing Year and Fuel Type



Demo Link :-<https://github.com/Harshit07-ux-dev/Car-Market-Trends-Analysis-with-Car-Dekho-Data->

Thank you